

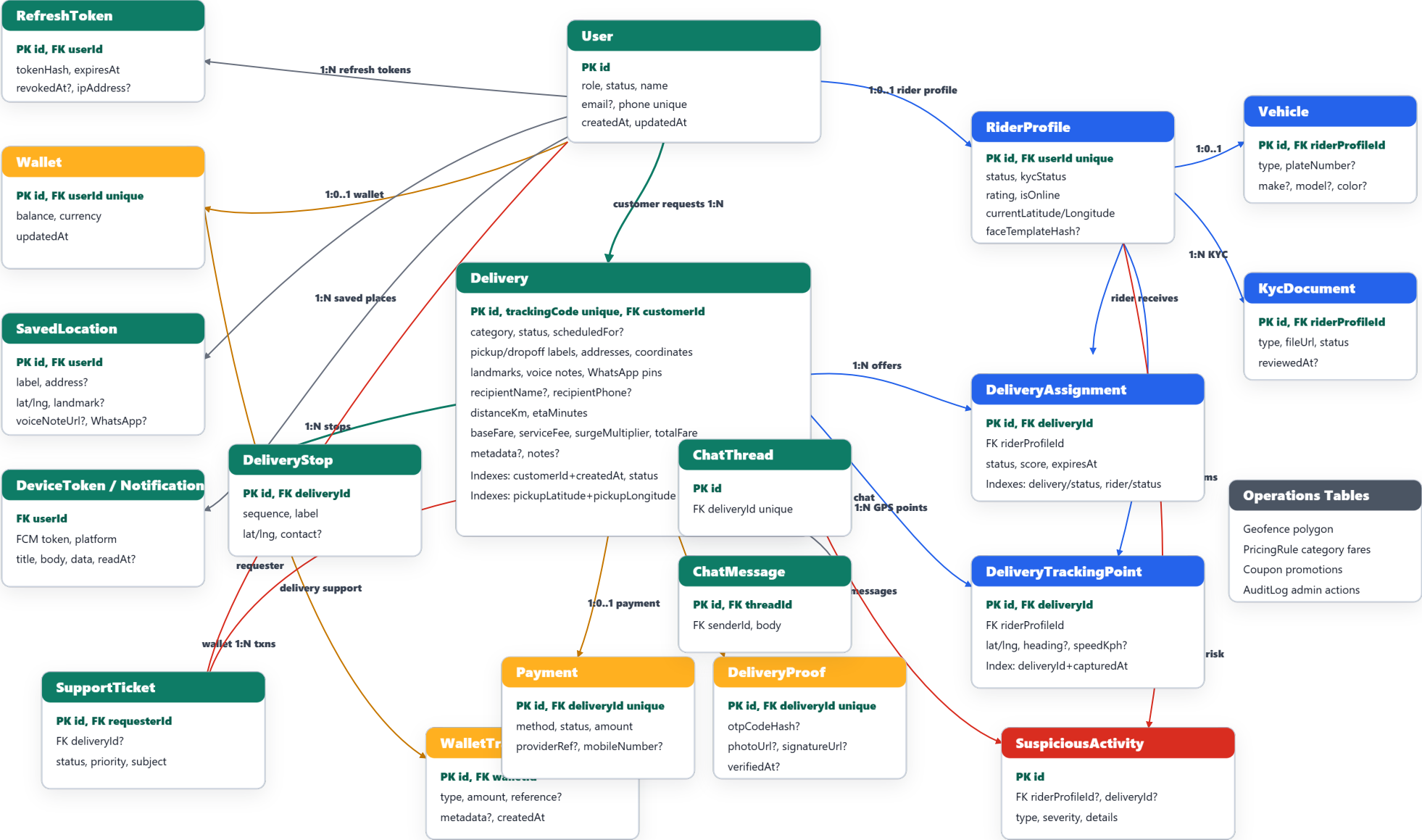
Benbax Go Entity Relationship Diagram

Production database design for customer delivery, rider dispatch, real-time tracking, payments, wallet, support, geofencing, fraud detection, and admin operations.

Source apps/api/prisma/schema.prisma
Database PostgreSQL via Prisma ORM
Prepared May 7, 2026

Logical ERD

Customer/Core Rider/Dispatch Payments Risk/Safety



Entity Catalog And Relationship Matrix

The schema is optimized around fast delivery creation, rider matching, live GPS, payment reconciliation, customer support, and operational control.

Cardinality 1:1, 1:0..1, 1:N
Delete model Mostly cascade on dependent records
Scale path PostGIS and tracking partitions

Identity and access

User, RefreshToken, DeviceToken, Notification, and AuditLog support customer, rider, admin, operations, and support roles.

Delivery lifecycle

Delivery owns stops, assignments, tracking points, payment, proof, chat, support links, emergency incidents, and risk flags.

Rider operations

RiderProfile links a user to vehicle, KYC documents, availability, location, assignments, tracking points, and suspicious activity.

Money movement

Payment captures delivery charges. Wallet and WalletTransaction handle customer balances, refunds, rider payouts, and incentive accounting.

Primary Relationships

PARENT	CHILD	CARDINALITY	PURPOSE
User	Delivery	1:N	A customer can request many deliveries. Each delivery has one customer.
User	RiderProfile	1:0..1	A rider account extends the base user profile with KYC, location, and dispatch data.
RiderProfile	DeliveryAssignment	1:N	Riders receive dispatch offers with score, response state, and expiry.
Delivery	DeliveryAssignment	1:N	A delivery can have multiple offers over time until one is accepted.
Delivery	DeliveryTrackingPoint	1:N	Live GPS stream and historical route replay.
RiderProfile	DeliveryTrackingPoint	1:N	Each tracking point belongs to the rider who produced it.
Delivery	Payment	1:0..1	One active charge record per delivery for MoMo, Paystack, wallet, or COD.
Delivery	DeliveryProof	1:0..1	OTP, proof photo, signature, and final verification metadata.
User	SavedLocation	1:N	Localized Ghana addresses with landmarks, voice notes, and WhatsApp pins.
Delivery	SupportTicket / EmergencyIncident / SuspiciousActivity	1:N	Operational controls for complaints, emergencies, and fraud investigation.

Production recommendation: enable PostGIS for geospatial matching and geofence containment, then partition DeliveryTrackingPoint by time as tracking volume grows.